

Def.

Let G be a group and A be a non-empty subset. A map

$$\alpha: G \times A \rightarrow A: (g, a) \mapsto g \cdot a$$

is called a group action of G on A if it satisfies

- 1) For all $a \in A$, $1_G \cdot a = a$
- 2) For all $g_1, g_2 \in G$, and $a \in A$, $(g_1 g_2) \cdot a = g_1 \cdot (g_2 \cdot a)$

Prop. Given a group action α of G on A , there exists a homomorphism

$$\mathcal{Q}_\alpha: G \rightarrow S_A: g \mapsto \sigma_g \quad \text{where} \quad \sigma_g: A \rightarrow A: a \mapsto g \cdot a.$$

which is called the permutation representation associated to the given action.

Conversely, given a homomorphism $\mathcal{Q}: G \rightarrow S_A$, there exists a group action

$$\alpha_{\mathcal{Q}}: G \times A \rightarrow A: (g, a) \mapsto \mathcal{Q}(g)(a)$$

Proof. We only need to check that the given map is well-defined,

Drumt, Sec. 4.1.

Thmmt. sec 4.2.

